



Comparison of interatomic potential models on the molecular dynamics simulation of fast-ion conductors: A case study of a Li garnet oxide $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$

Jin Dai^a, Qian Chen^a, Tobias Glossmann^{b,c}, Wei Lai^{a,*}

^a Department of Chemical Engineering and Materials Science, Michigan State University, East Lansing, MI 48824, USA

^b Mercedes-Benz Research & Development North America, Inc., 12120 Telegraph Road, Redford, MI 48239, USA

^c Department of Chemistry, Oakland University, Rochester, MI 48309, USA

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ABSTRACT

In this work, we performed molecular dynamic simulations of a representative fast-ion conductor ($\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$) using three different interatomic potential models: non-polarizable core (Core), polarizable core-shell (CS), and induced dipole (ID). To compare these three models, we calculated the phase transition behavior, self-diffusivity, ionic conductivity, dielectric constant, and elastic moduli of $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ and checked results against experimental and other simulation work. We found that overall all three potential models predict results that agree reasonably well with experiments and first-principles calculations. However, we think that the ID model that combines both force-matching, atomic polarization, and first-principles charge has the technical advantage over the other two, while exhibiting slightly higher computational cost. We also discussed ways to further improve these interatomic potential models.

1. Introduction

Among solid electrolytes, Li-ion conducting garnet-type oxides have become promising candidates for all-solid-state Li-ion batteries due to their high ionic conductivity and stability against elemental lithium [1,2]. Since Murugan et al. synthesized highly conductive cubic garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO) in 2007 [3], substantial studies have been done to understand its crystal structure and mechanism of Li ion conduction [4–7]. Although LLZO was first identified as a cubic structure with space group Ia3d at room temperature, it was verified that undoped LLZO has a tetragonal phase with space group I4₁/acd [6]. Tetragonal LLZO (t-LLZO) shows a lower ionic conductivity by about 2 orders of magnitude than cubic LLZO (c-LLZO), which is attributed to its complete ordering of Li atoms and vacancies.

Classical molecular dynamics (MD) simulations have been widely used in the research of physical and chemical properties of materials including solid electrolytes. The reliability of classical MD simulations highly depends on the potential used to represent the interatomic interactions. The interaction between the particles are given by empirical potential functions comprising a set of parameters. Fig. 1 presents the schematics of four models that have been commonly used in classic MD simulations. The non-polarizable model describes the atom as a single

core with a fixed charge. Cieplak et al. [8] suggested that interatomic potentials become more accurate by adding polarization effects and there are three polarizable models. First, the polarizable fluctuating charge model considers that the atomic charge changes with the local environment and generally applies the principle of electronegativity equalization to obtain the dynamic charge. Second, the polarizable core-shell (also known as Drude oscillator) model describes an atom as a core plus a shell particle connected by a spring, with shell representing electronic polarization. Third, the polarizable induced dipole model describes an atom as a point charge plus a dipole (induced electronic polarization).

In the application of these models to the study of garnets, Adams et al. [9] performed MD simulation on LLZO with the core model and Morse potential. Chen et al. [10] also used the core model to study the diffusion characteristics of LLZO based on the Buckingham potential. In our previous work [5,11], we used the core model to study the size effect on MD simulation of LLZO and used polarizable core-shell model to study the local structure and dynamics of LLZO. Burbano et al. [12] used the polarizable induced dipole model to perform MD simulation of LLZO to study the mechanism of its low ionic conductivity. However, no work has been done to compare these three models.

In this paper, we compared results of classic MD simulation on LLZO

* Corresponding author.

E-mail address: laiwei@msu.edu (W. Lai).

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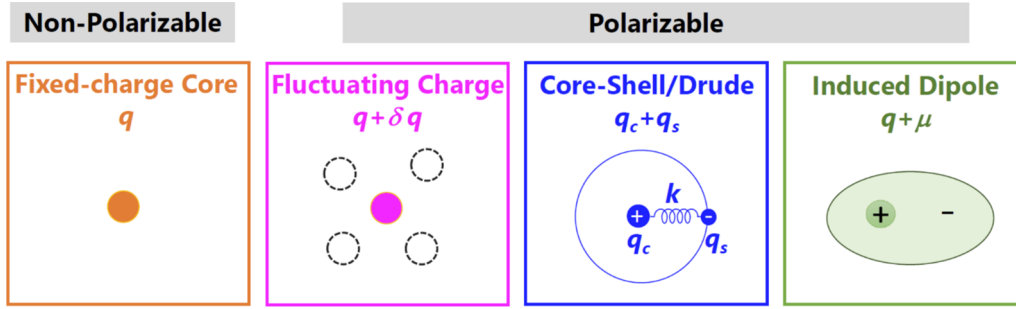


Fig. 1. Schematics of potential models used in MD simulation which includes the non-polarizable fixed-charge core model that describes the atom as a single core with a fixed charge and three different polarizable models, i.e., fluctuating charge, core-shell, and induced dipole models.

using three models, i.e., non-polarizable core (Core) model, polarizable core-shell (CS) model, and induced dipole (ID) model. In CS model, we only assigned O as the polarizable atom. In ID model, we assigned the electronic polarizability to O, La, and Zr species. We compared the accuracy of these three models in reproducing the phase transition, self-diffusivity, ionic conductivity, dielectric constant, and elastic moduli of fast-ion conducting LLZO.

2. Methods

2.1. Core and core-shell (CS) models

MD simulations were performed with the DL_POLY Classic package for both Core and CS models [13]. A $2 \times 2 \times 2$ supercell of $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ was used with 1536 atoms in the Core model and 2304 atoms (including O shell) in the CS model. In the Core model, the charges were using the DDEC (Density Derived Electronic and Chemical) approach [14–16] from the result of density-functional theory calculations. Ionic interactions include the long-range Coulombic potential and short-range Buckingham potential. Potential parameters were identical to those in our previous studies [11] and were listed in Table 1. In the CS model, the Dick-Overhauser [17] core-shell polarizable potential were added for O atom with the O shell (OS) mass set to 0.2 a.u., as in our previous studies [5,18]. However, the charges of each species were set as 80% of their formal charges and we added the Morse potential to account for possible covalent interaction. Potential parameters for the CS model were obtained by fitting to experimental crystal structures (lattice parameters and atomic positions) and summarized in Table 2.

The velocity-Verlet integration was employed with a time step of 1 fs for the Core model and 0.25 fs for the CS model. The constant number, stress, and temperature (N σ T) ensemble (Berendsen thermostat and barostat relaxation times were set to be 0.1 and 0.5 ps, respectively) simulations were performed for 75 ps and the last 35 ps were averaged to obtain the lattice parameters. The equilibration time of constant number, volume, and energy (NVEe) ensemble was set to be 100 ps with the atomic velocities rescaled every 0.1 ps. Subsequent NVE ensemble simulations were performed to extract the

Table 1

Interatomic potential parameters for the Core model. The Buckingham interaction is described by $U_{ij}^{\text{Buckingham}} = A \exp(-r_{ij}/\rho) - C/r_{ij}^6$, where r_{ij} is the separation of ion i and j , and A , ρ , C the interatomic potential parameters.

species	q (e)	Buckingham pair interaction with O		
		A (eV)	ρ (Å)	C (eV Å ⁶)
La	2.50	2075.26	0.326	23.25
Li	1.00	1087.29	0.260	0.00
O	−1.65	4870.00	0.267	77.00
Zr	2.65	1650.32	0.311	5.10

diffusion and conduction behavior as well as the elastic moduli. The production time was set from 500 ps at high temperatures and to 70 ns in the case of low mobility of Li^+ in tetragonal phase at 800 K.

2.2. Induced-dipole (ID) model

In the ID model, atomic point charges were obtained from first-principles and ionic interactions include the electrostatic (point charge with point charge, charge with dipole, and dipole with dipole) and short-range Buckingham potentials. Potential parameters were derived from the force-matching method [19] which fits interatomic potential forces to first-principles forces of each atom.

Briefly, first-principles molecular dynamics based on the density-functional theory simulations were performed using the generalized gradient approximation (GGA-PBESol) [20] and the projector augmented wave (PAW) [21] method as implemented in the Vienna Ab Initio Simulation Package (VASP) [22]. The unit cell was simulated using the NPT ensemble at 1200 K. The energy cutoff of 500 eV was used for the plane-wave and the Brillouin zone was sampled at the Γ -point only. The production time was 3 ps with a time step of 1 fs. The trajectories after the first 0.15 ps were used to perform the force-matching. We extracted 190 configurations, i.e. one configuration every 15 fs. The obtained average structure was used for the subsequent static calculation. The average charge for each atom species was obtained using the DDEC approach.

The force-matching and MD simulations of the ID model were carried out using the FIST code as implemented in cp2k [23,24]. The charge of each species and potential parameters are listed in Table 3. The MD simulations were performed in a $2 \times 2 \times 2$ supercell with 1536 atoms. The constant number, pressure and temperature (NPT) ensemble was performed for 50 ps and the average lattice parameters were extracted from the last 40 ps. The Nose thermostat and barostat relaxation times were set to be 0.05 and 0.25 ps, respectively. The NVE equilibrium (NVEe) process was 50 ps and the maximum accepted deviation of the temperature from the desired target temperature was 50 K. The NVE process was set from 200 ps to 10 ns for different temperatures. The time step was 1 fs for NPT, NVEe and NVE process.

$U_{ij}^{\text{pol}} = \sum_i \frac{1}{2\alpha_i} |\mu_i|^2 + \sum_{ij} [(q_i \mu_j^\alpha g_{ij}(\mathbf{r}_{ij}) - q_j \mu_i^\alpha g_{ji}(\mathbf{r}_{ij})) T_{ij}^\alpha - \mu_i^\alpha \mu_j^\beta T_{ij}^{\alpha\beta}]$, where α_i is isotropic point polarizability of atom i , μ_i the induced dipole moment at the i th site, which is treated as additional degrees of freedom and obtained at each MD step by minimizing the polarization energy. T_{ij} is the dipole-dipole interaction tensor. The short-range correction to atom i and atom j was in the form of Tang-Toennies damping function as [25] $g_{ij}(\mathbf{r}_{ij}) = 1 - c_{ij} e^{-b_{ij} r_{ij}} \sum_{k=0}^4 \frac{(b_{ij} r_{ij})^k}{k!}$.

2.3. Diffusion: Incoherent density correlation function

Atomic diffusion was investigated by the species-specific incoherent density correlation function [26] which was calculated according to $I_{inc}^\alpha(Q, t) = \left\langle \frac{1}{N_\alpha} \sum_{l=1}^{N_\alpha} e^{-iQ \cdot \mathbf{r}_l^\alpha(0)} e^{iQ \cdot \mathbf{r}_l^\alpha(t)} \right\rangle$, where α represents lithium

Table 2

Interatomic potential parameters for the CS model in which oxygen shell is presented as OS. The Morse interaction is described by $U_{ij}^{\text{Morse}} = E_0 [1 - \exp(-k(r_{ij} - r_0))]^2 - 1$, where r_{ij} is the separation of ions i and j , and E_0 , k , r_0 the potential parameters. The harmonic spring constant between oxygen core and shell is k_{cs} .

species	q (e)	Interaction pair	Buckingham potential			Morse potential			Core-shell potential for O
			A (eV)	ρ (Å)	C (eV Å ⁶)	E_0 (eV)	k (Å ⁻¹)	r_0 (Å)	k_{cs} (eV Å ⁻²)
La	2.4	La-OS	1828	0.328	0	0.066	1.471	2.423	19.64
Zr	3.2	Zr-OS	2135	0.302	0	0.241	2.742	2.016	
Li	0.8	Li-OS	370	0.291	0	0.081	0.870	1.937	
O	-2.116	Li-Li	1470	0.239	0				
OS	0.516								

species, N_α the number of lithium atoms, and $\mathbf{r}_l^\alpha(t)$ the position of l th atom at time t . $\mathbf{Q}$ is the wave vector transfer as $\mathbf{Q}_{n_x n_y n_z} = n_x \mathbf{a}^* + n_y \mathbf{b}^* + n_z \mathbf{c}^*$, where $\mathbf{a}^*$, $\mathbf{b}^*$, $\mathbf{c}^*$ are reciprocal unit vectors and n_x , n_y , and n_z are all integers. Ignoring data for the first 5 ps that undergo short-time dynamics, we fit the $I_{\text{inc}}^\alpha(Q, t)$ curve using a stretched exponential (Kohlrausch-Williams-Watts) function [27] $I^{\text{KWW}}(Q, t) = \exp(-[\Gamma^{\text{KWW}}(Q)t]^\beta)$ where β is the stretching parameter. The mean relaxation rate $\langle \Gamma \rangle$ was obtained from $\Gamma^{\text{KWW}}(Q)\beta/G(\beta^{-1})$, where G is the Gamma function. After obtaining the plot of Γ as a function of Q^2 , the Fickian diffusion model $\Gamma(Q) = DQ^2$ was used to fit the first three Q values since Γ is linear dependent of Q^2 at smaller Q corresponding to the long-range diffusion, where D is the self-diffusivity. The diffusion of Li at short range corresponding to larger Q values was described by the Singwi-Sjölander (SS) jump diffusion model [28,29] with the formula $\Gamma^{\text{SS}}(Q) = \frac{1}{\tau} \frac{Q^2 \langle r^2 \rangle / 6}{1 + Q^2 \langle r^2 \rangle / 6}$, where τ is the residence time and $\langle r^2 \rangle$ the mean square jump length. This model assumes that the time Li atoms taken for diffusion is much shorter than the residence time of Li atoms at a site.

2.4. Conduction: Coherent charge current density correlation function

Ionic conduction was investigated by the coherent charge current density correlation function [26] that includes two components: the longitudinal component

$C^{L,c}(Q, t) = \frac{e^2}{V k_B T} \frac{1}{Q^2} \langle (\mathbf{Q} \cdot \mathbf{J}(Q, 0))(\mathbf{Q} \cdot \mathbf{J}(-Q, t)) \rangle$, and the transverse component $C^{T,c}(Q, t) = \frac{e^2}{V k_B T} \frac{1}{2Q^2} \langle (\mathbf{Q} \times \mathbf{J}(Q, 0)) \cdot (\mathbf{Q} \times \mathbf{J}(-Q, t)) \rangle$, where $\mathbf{J}(Q, t) = \sum_{n=1}^N q_n \mathbf{v}_n(t) e^{-i\mathbf{Q} \cdot \mathbf{r}_n(t)}$ is the collective charge current, e the elementary electron charge, V the volume of the simulation cell, k_B the Boltzmann constant, T the temperature, q_n the charge of the n th atom, and $\mathbf{v}_n$ the velocity of the n th atom. The Fourier Transform (FT) of longitudinal and transverse forms are $S^{L,T}(Q, \omega) = \int_0^\infty C^{L,T}(Q, t) e^{i\omega t} dt$, where ω is the frequency.

2.5. Elastic moduli: Momentum correlation function

The elastic moduli were investigated by the momentum correlations that can be separated into the longitudinal and transverse components as

Table 3

Interatomic potential parameters for the ID model.

Species	q (e)	Polarization potential				Buckingham pair interaction with O		
		α (Å ³)	Ion-pair	b_{ij} (Å ⁻¹)	c_{ij} (Å ⁻¹)	A (eV)	ρ (Å)	C (eV Å ⁶)
La	2.54	1.71	La-O	3.344	2.315	2512.92	0.314	0
Zr	2.93	0.76	Zr-O	3.155	1.233	3034.17	0.282	0
Li	1	-	Li-O	3.772	1.729	507.79	0.298	0
O	-1.71	1.74						

The polarization potential is described by [25].

$C^{L,m}(Q, t) = \frac{1}{Q^2} \langle (\mathbf{Q} \cdot \mathbf{J}(Q, 0))(\mathbf{Q} \cdot \mathbf{J}(-Q, t)) \rangle$, $C^{T,m}(Q, t) = \frac{1}{2Q^2} \langle (\mathbf{Q} \times \mathbf{J}(Q, 0)) \cdot (\mathbf{Q} \times \mathbf{J}(-Q, t)) \rangle$ where $\mathbf{J}(Q, t) = \sum_{n=1}^N m_n \mathbf{v}_n e^{-i\mathbf{Q} \cdot \mathbf{r}_n(t)}$ is the collective momentum, m_n the atomic mass of the n th atom, $\mathbf{v}_n$ the velocity of the n th atom. The Fourier Transform (FT) of longitudinal and transverse forms are $S^{L,T}(Q, \omega) = \int_0^\infty C^{L,T}(Q, t) e^{i\omega t} dt$, where ω is the frequency.

We located the peaks (ω) of $S^{L,T}(Q, \omega)$ and assumed linear dispersion of ω on Q to obtain longitudinal and transverse sound velocities v_L and v_T . For the cubic phase, the elastic constants C_{11} , C_{44} and C_{12} were calculated by the following formulas [30]:

$$(v_L^2 \rho)_{[100]} = C_{11}, (v_T^2 \rho)_{[100]} = C_{44}, (v_T^2 \rho)_{[110]} = \frac{1}{2}(C_{11} - C_{12})$$

For the tetragonal phase, C_{11} , C_{44} and C_{12} were calculated in the same way as in the cubic phase, while C_{33} and C_{66} were calculated by the following formulas [30]:

$$(v_L^2 \rho)_{[001]} = C_{33}, (v_T^2 \rho)_{[100]} = C_{66}$$

We assumed that $C_{12} = C_{23}$ as they differ by only 2% from Deng et al. [31].

The bulk modulus, shear modulus and Poisson's ratio were calculated according to

$$B = \frac{(C_{11} + C_{22} + C_{33}) + 2(C_{12} + C_{23} + C_{31})}{9}$$

$$G = \frac{(C_{11} + C_{22} + C_{33}) - (C_{12} + C_{23} + C_{31}) + 3(C_{44} + C_{55} + C_{66})}{15}$$

$$\nu = \frac{3B - 2G}{6B + 2G}$$

In the cubic phase, $C_{11} = C_{22} = C_{33}$, $C_{12} = C_{23} = C_{31}$, $C_{44} = C_{55} = C_{66}$. In tetragonal phase, $C_{11} = C_{22}$, $C_{44} = C_{55}$, $C_{23} = C_{31}$.

3. Results

3.1. Lattice parameters

First we checked if all three potential models could reproduce the tetragonal to cubic phase transition of LLZO observed experimentally. A wide variety of transition temperatures were reported for LLZO experimentally ranging from 373 to 923 K [4,7,32–34]. The difference

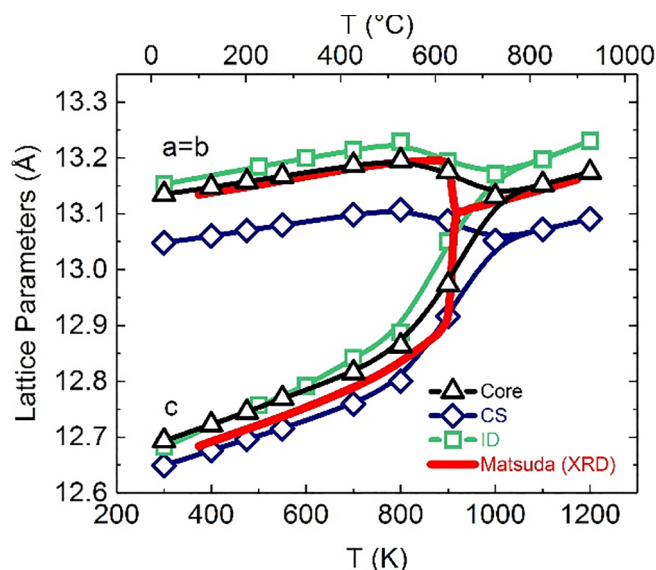


Fig. 2. Lattice parameters from Core, CS and ID model as a function of temperature. XRD experimental values from Matsuda et al. [33] are shown for comparison.

could be resulting from the impurity / dopant (i.e., Al^{3+}) incorporation and/or H_2O or CO_2 exposure during material synthesis and characterization. For example, Matsuda et al. [33] found that phase transition temperature of LLZO decreases as the Al content increases and the reversible phase transition between the tetragonal phase and cubic phase occurs at 640°C (913 K) for pure LLZO without Al. Larraz et al. [35] reported that the presence of H_2O could induce the tetragonal to cubic phase transition due to the H^+/Li^+ exchange mechanism. On the other hand, undoped LLZO not exposed to humidity at any moment undergoes a reversible tetragonal to cubic phase transition at about 645°C (918 K). Given these observations, we only plotted the experimental values from Matsuda et al. [33] that applies to pure LLZO in Fig. 2, along with lattice parameters obtained from the Core, CS and ID models. Lattice parameters of all three models increase with increasing temperature as expected in both tetragonal and cubic phases, with the phase transition occurring at about 900 to 1000 K. Both Core and ID models show thermal expansion and phase transition behaviors consistent with experiments while the CS model shows lower lattice parameters in *a*.

3.2. Self-diffusivity, jump length and residence time

Next we compared the long-range diffusion parameter, i.e. self-diffusivity, and short-range parameters such as jump length and residence time. The residence time and jump length obtained from the SS model are shown in Fig. 3. All Core, CS and ID models show a sharp increase of residence time (Fig. 3a) at the phase transition temperature. CS and ID models have close values in the cubic phase. Activation energies obtained from an Arrhenius fit for Core, CS and ID models are 0.26, 0.26 and 0.31 eV, respectively. In the tetragonal phase, all three models show high activation energy values of above 1.7 eV. The mean jump length (Fig. 3b) obtained from Core, CS and ID models at higher temperatures (cubic phase) are very close, i.e., 1.4–1.7 Å, which is roughly the distance between 24d (tetrahedral) and 48g (octahedral) sites in the cubic phase [5]. The jump length exhibit a significant increase at phase transition temperature. In the tetragonal phase (800 K), the jump lengths for CS and ID models are about 2–2.5 Å, which is roughly the distance between 8a (tetrahedral) and 32g (octahedral) sites [5,6].

Fig. 4 shows the self-diffusivity of Li^+ in LLZO for the Core, CS, and

ID models. Our calculated values from the Core model are close to those from Jalem et al. [36] using first-principles molecular dynamics (FPMD) simulation with $1 \times 1 \times 1$ cell size, while values from the CS and ID models are a few times higher. The activation energy of Core model for the cubic phase is 0.24 eV, which is also close to that from Jalem et al. [36] at the temperature range of 873 to 1473 K, 0.26 eV. The activation energy of CS and ID models for cubic phase are 0.18 and 0.20 eV, respectively. The self-diffusivity obtained from the CS model at high temperatures is close to that from Burbano et al. also using an ID model but with a smaller activation energy of 0.14 eV. The difference is likely to result from the lower phase transition temperature of 623 K, as opposed to 900–1000 K, in their work. Experimentally, the Li self-diffusivity at ambient temperature from Kuhn et al. using the spin-lattice relation Nuclear Magnetic Resonance (NMR) measurement is on the order of $10^{-14} \text{ cm}^2\text{s}^{-1}$. Although our simulation was unable to calculate Li self-diffusivity at room temperature, extrapolation of our data to the room temperature yields values on the order of 10^{-25} – $10^{-20} \text{ cm}^2\text{s}^{-1}$, which are much lower than the experimental result.

3.3. Ionic conductivity and dielectric constant

Ionic conductivity is one of the most important properties of fast-ion conductors and it has been widely studied for both tetragonal and cubic LLZO. Here, we calculated the ionic conductivities of LLZO using the method similar to our previous study [11] to compare the Core, CS and ID models. Fig. 5(a) gives an example of the transverse component of coherent charge current density correlation function $C^{T,c}(\mathbf{Q}, t)$ of the ID model at 1000 K for three smallest $\mathbf{Q}$. It was found that the Fourier transform of longitudinal component is a peak that goes to zero at zero frequency (not shown) but the transverse component, $S^T(\mathbf{Q}, \omega)$ shown in Fig. 5(b), has a non-zero intercept suggesting that LLZO is ionic conducting material. The ionic conductivity was extrapolated from the transverse component $\sigma = \lim_{\omega \rightarrow 0} S^T(\mathbf{Q}, \omega)$. To manifest that $S^T(\mathbf{Q}, \omega)$ is $\mathbf{Q}$ independent, it is important to separate out all transverse outputs as shown in Fig. 5(c). For example, the transverse polarization of vector [1 0 0] has two directions, i.e., [0 1 0] and [0 0 1]. It is apparent that outputs of the two $\mathbf{Q}$ vectors are similar to each other, which indicates that $S^T(\mathbf{Q}, \omega)$ is independent of $\mathbf{Q}$ vector. Thus we averaged all six values of the first $\mathbf{Q}$ vector and took the result of the smallest frequency as the ionic conductivity.

Fig. 6 summarizes the ionic conductivities for Core, CS and ID models at different temperatures. It is obvious that all three models have a kink at about 900 K corresponding to the phase transition temperature. The experimental measured ionic conductivities from Wang et al. [4] and Matsui et al. [37] also exhibit a sharp decrease at phase transition temperature between 600°C (873 K) and 650°C (925 K). Activation energies of the cubic phase for Core, CS and ID models, i.e., Core-c, CS-c, ID-c, are 0.28, 0.13 and 0.17 eV, respectively, while values from Wang et al. [4] and Matsui et al. [37] are 0.13 and 0.12 eV, respectively. For the tetragonal phase, our simulations are unable to reliably calculate the ionic conductivities at lower temperatures due to the low Li^+ mobility. A simple extrapolation of our data will yield lower values to those from Wang et al. [4], Matsui et al. [37], and J. Awaka et al. [6].

Since the conductivity is related to dielectric function as $\omega\epsilon_0[\epsilon_{L,T}(\mathbf{Q}, \omega) - 1] = i\sigma_{L,T}(\mathbf{Q}, \omega)$, the real part of dielectric function can be obtained through the imaginary part of conductivity as $\text{Re}[\epsilon_T(\mathbf{Q}, \omega) - 1] = -\text{Im}[\sigma_T(\mathbf{Q}, \omega)]/(\omega\epsilon_0)$. Fig. 7(a) gives an example of the imaginary part of transverse conductivity $\sigma_T(\mathbf{Q}, \omega)$ over ϵ_0 for the first $\mathbf{Q}$ value ($0.24 \text{ \AA}^{-1}$) as a function of frequency for the ID model at 800 K. The linear fit (inset of Fig. 7(a)) yields a slope of $\epsilon_r(0) - 1$ where $\epsilon_r(0)$ is the static (zero frequency) dielectric constant. The results of $\epsilon_r(0)$ of all three models at different temperatures are summarized in

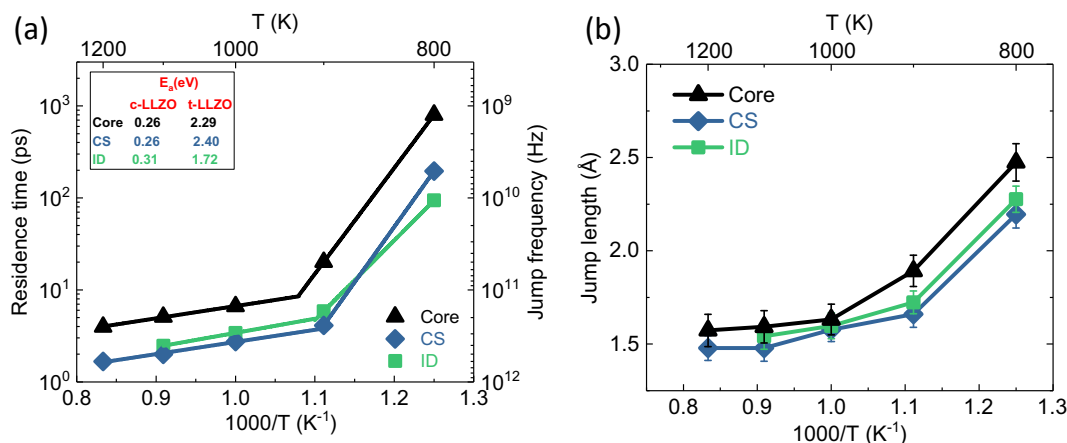


Fig. 3. (a) Mean residence time (τ) of Li diffusion in LLZO obtained from Core, CS, and ID models in both cubic (c) and tetragonal (t) phases. Solid lines are an Arrhenius fit. (b) Mean jump length of Li diffusion in LLZO obtained from Core, CS, and ID models.

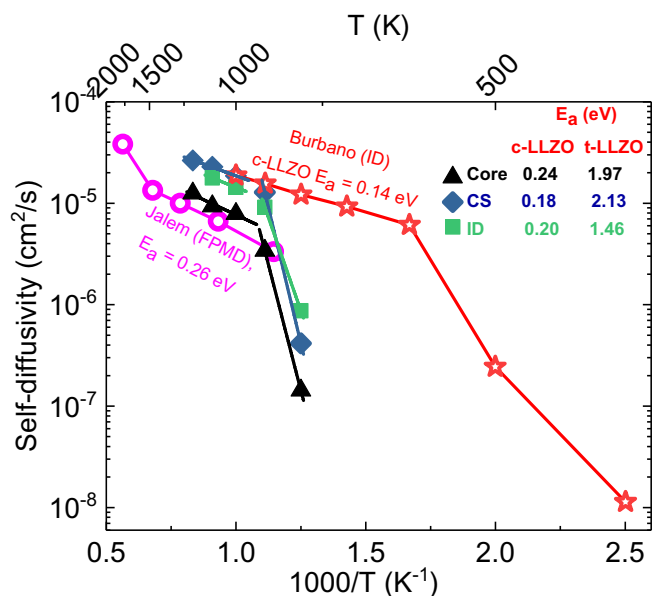


Fig. 4. Self-diffusivity as a function of temperature for Core, CS, and ID models in both cubic (c) and tetragonal (t) phases. Solid lines are Arrhenius fit. The FPMD simulation data by Jalem et al. [36] and classic MD simulation data by Burbano et al. [12] are shown for comparison.

Fig. 7(b). In the tetragonal phase, dielectric constants of three models decreases with increasing temperatures, as expected for typical materials due to thermal agitation. On the other hand, dielectric constants of three models increases with increasing temperatures and we believe this is due to mobile Li ions. The dielectric constant of tetragonal and cubic LLZO is not known but S. Narayanan et al. [38] reported that $Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$ ($x = 0.25, 0.5$ and 0.75) have dielectric constants around 60 below room temperature.

3.4. Bulk modulus (B) and shear modulus (G)

Finally, elastic moduli values from the Core, CS and ID models were presented. Fig. 8 shows the bulk modulus (B) and shear modulus (G) for three models. Both B and G decreases with increasing temperature, suggesting the material softens upon heating. The Core model has the highest values of both B and G, while the CS and ID models have similar elastic moduli. Experimental values at 300 K from Ni et al. [39] are presented in Fig. 8 and they are similar to our computational values.

4. Discussions and conclusions

In this work, we compared one non-polarizable (core/Core) and two polarizable (core-shell/CS and induced-dipole/ID) models for the molecular dynamics simulation of a fast-ion conductor, $Li_7La_3Zr_2O_{12}$ (LLZO). Overall, we think it is difficult to explicitly point out which model is the best, as all three models can reproduce various structural

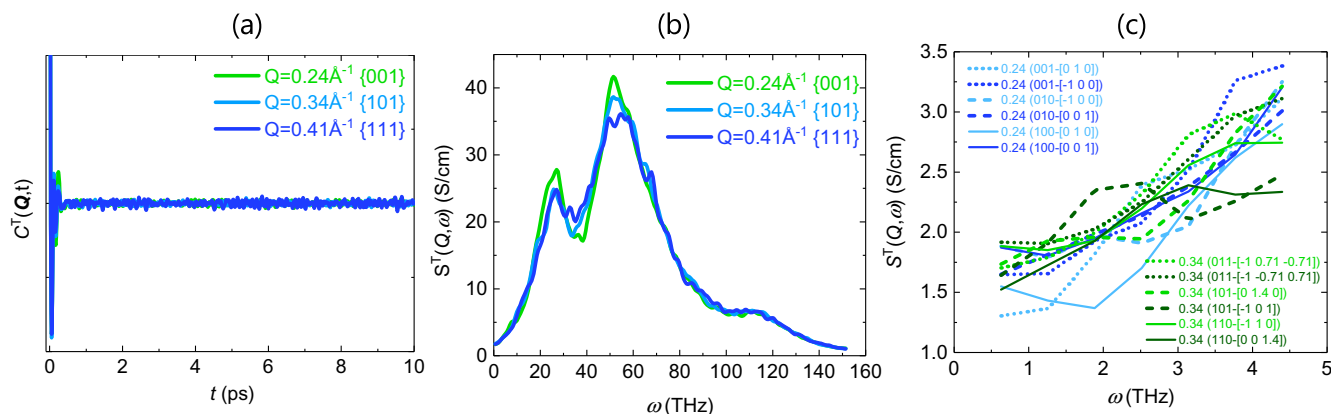


Fig. 5. (a) Time and (b) frequency domain of the transverse coherent charge current density correlation function for the first three Q values of the ID model at 1000 K. (c) All transverse outputs for first two Q vectors at low frequencies. The first number in the legend denotes the Q value in Å^{-1} while the second and third series of three digits denote the Q direction and its transverse direction.

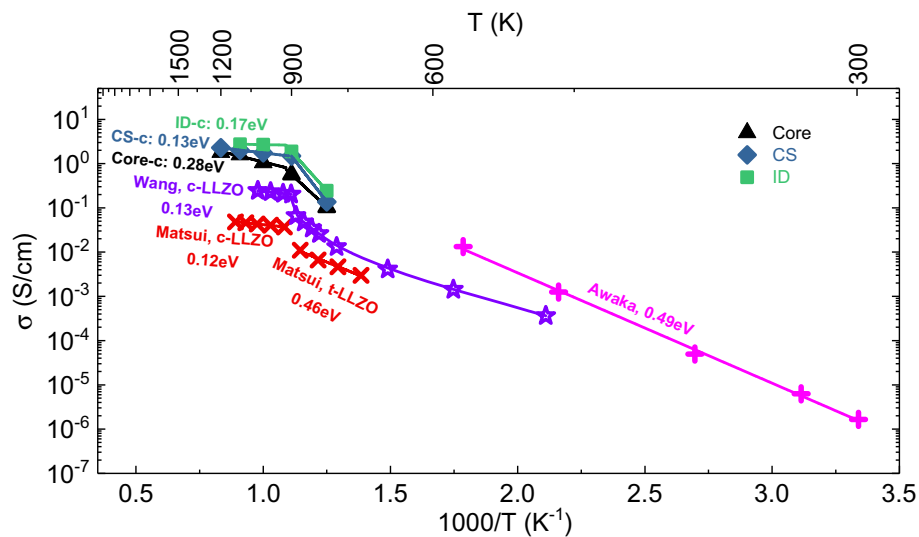


Fig. 6. Ionic conductivity as a function of temperature for Core, CS, and ID models. Literature data from experimental measurements by Wang et al. [4], Matsui et al. [37], and Awaka et al. [6] are shown for comparison.

and dynamic properties (lattice parameters, diffusivity, conductivity, dielectric constants, and elastic moduli) of LLZO reasonably well. Due to various limitations of existing codes, we used different methods to obtain potential parameters, e.g. arbitrary vs first-principles charge (note that the charges of each species for all three models were partial charges so they are customized to each material), Buckingham vs Buckingham + Morse short-range interaction, empirical fit vs force-matching, etc. We believe this also complicates the direct comparison of different potential models. On the other hand, we think the ID model combined with the force-matching approach is the most advanced one as it includes the polarization through atomic polarizability and takes from first-principles results as the input. The CS model includes the polarization through a spring-connected shell where parameters of shell charge and spring constant are almost entirely empirical. Looking forward, we think that the ID model can be improved by incorporating the anisotropy in the atomic polarizability and fluctuating charges, where the charge will vary with the local chemical environment. Furthermore, only pair potentials were considered in these three models while many-body potentials will be more accurate to describe the mixed metallic/covalent nature of the bonding.

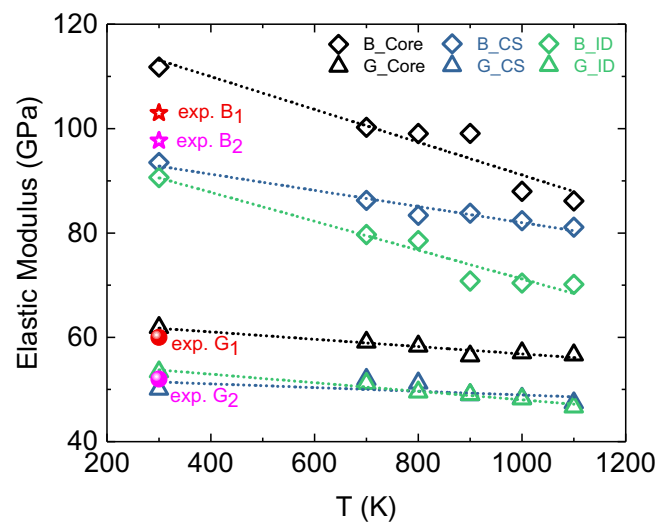


Fig. 8. Elastic modulus as a function of temperature for Core, CS and ID models. Experimental values from Ni et al. [39] are shown for comparison.

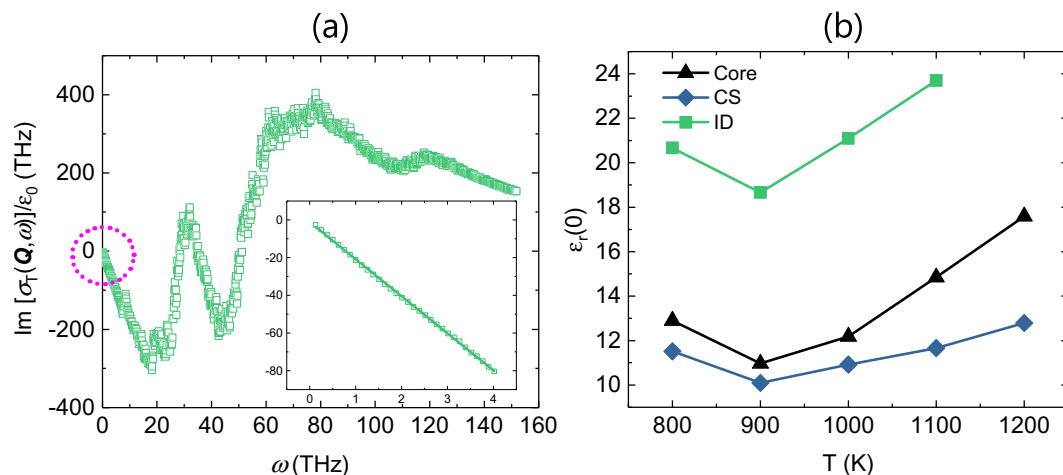


Fig. 7. (a) The imaginary part of transverse conductivity $\sigma_T(Q, \omega)$ over ϵ_0 as a function of frequency for the ID model at 800 K. The inset shows details at low frequencies and the solid lines is linear fit. (b) The dielectric constant at zero frequency $\epsilon_r(0)$ as a function of temperature for Core, CS, and ID models.

CRediT authorship contribution statement

Jin Dai: Data curation, Formal analysis, Writing - original draft.
Qian Chen: Data curation, Methodology.
Tobias Glossmann: Validation, Methodology.
Wei Lai: Conceptualization, Formal analysis, Methodology, Writing - review & editing.

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Data Availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

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